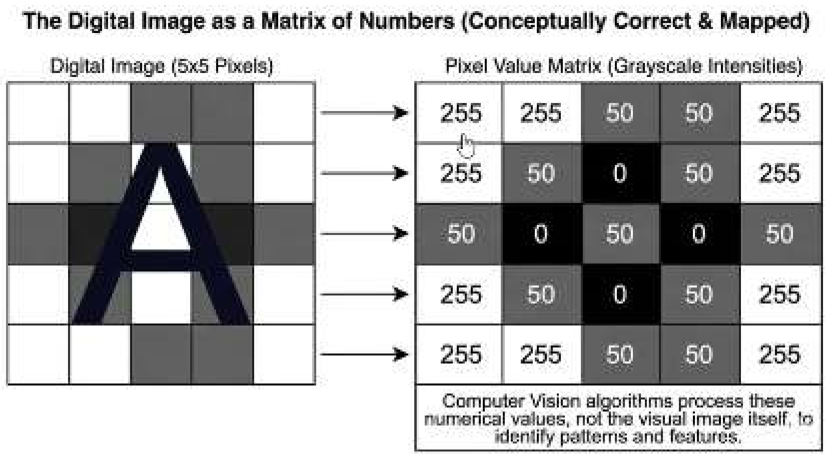
DAY – 1

Topics - Computer Vision Overview , AI vs ML vs Deep Learning vs Generative AI , Real World CV use Cases and training outcomes

Computer Vision Overview :

- Computer vision is how an AI sees the world
- It's a way to analyse images return information about them and the objects in those images ,
- In simple terms , computer vision is capturing features of an image and helps with semantic understanding and helps performing various task like object detection , segmentation and generation.
- The computer doesn't sees image as human does as computer does see these images as pixels and numbers matrix
- It proceeses a grid of numerical values.



- How CV transforms pixels into understanding:
  - It takes pixels as input
  - Then can do obtain feature extraction like : Identifying edges ,shapes , textures and objects
  - Output :Semantic understanding :Classification , detection , segmentation .
- A single pixel is a number (or tuple)
- Grayscale images : One value from 0(black) to 255(white)
- Color(RGB) : Three values [R,G,B] , with three channels : R,G,B each of range 0-255 , defining the colors

➔ Computer Vision Pipelines : Input -> Process -> Output:

Stage	What It Is	Examples
The Input	The raw visual data provided to the system.	A single image, a video stream, a 3D point cloud from LiDAR, medical scan (MRI/CT).
The Process	Algorithms that **detect patterns** in the numerical data.	**Low-level:** Find edges, corners, blobs. **Mid-level:** Group features into shapes, textures. **High-level:** Recognize objects, scenes, activities.
The Output	The actionable information extracted.	A label ("cat"), bounding boxes, a segmented mask, a 3D reconstruction, a generated image, a control signal for a robot.

➔ Computer vision process practical outline :
Real-world scenario → Digital Sensors → Grid of pixels values (A 3D Tensor : Height X Width X Channels) → Computer Vision Algorithms finds patterns in numbers → Meaningful output

➔ We can use computer vision in various aspects of life :

$2^{8} = 256$ 0 - 255

Domain	Traditional Approach	Modern CV (Deep Learning)	Impact
Healthcare	Manual analysis by radiologists.	**AI detects tumors** in MRI scans.	Earlier, more accurate diagnosis.
Transportation	Basic lane sensors.	**Self-driving cars** perceive full 3D environment.	Safer, autonomous mobility.
Retail	Barcode scanners.	**Cashier-less stores**, shelf monitoring.	Frictionless shopping experience.
Security	Manual CCTV monitoring.	Automated anomaly & facial recognition.	Proactive public safety.

➔ Semantic Gaps : the core challenges of Computer vision :

Challenge	What Changes? (The Pixels)	What Stays the Same? (The Object)	Why It's Hard for a Computer
1. Viewpoint Variation	The entire 2D projection and visible features.	The 3D identity and structure of the object.	A **side-view** and **top-view** of a chair share almost no identical pixel patterns.
2. Illumination	Absolute RGB values, contrast, shadows.	The intrinsic color and material properties.	**A red apple in bright light vs. dim shadow** appears as two completely different sets of numbers.
3. Occlusion	A large portion of the object's pixels are missing, replaced by another object's pixels.	The identity of the partially visible object.	Must **recognize an object from a fraction of its usual features** (e.g., a cat by its tail alone).
4. Scale	The number of pixels occupied by the object in the image.	The semantic category of the object.	A small blob of pixels could be a distant car or a nearby toy; the system must use context to decide.

➔ Human Vision vs Computer Vision :

Capability	**Human Vision**	**Naive Computer Vision** (Pixel Matching)
Understanding	**Invariant.** Sees the **object** (apple, chair) despite pixel changes.	**Variant.** Sees only the specific **pixel pattern.** A different pattern is a different input.
Robustness	Handles viewpoint, light, occlusion, and scale **effortlessly** using lifelong 3D world experience.	**Fragile.** Fails completely without explicit algorithms designed for invariance.
Learning	Generalizes from few examples using powerful cognitive priors.	**Requires thousands of examples** showing all possible variations to learn the concept.

➔ In simple terms : Computer vision tries to bridge the profound disconnect between what a computer receives and what human perceives

Source (Udemy) : https://www.udemy.com/course/mastering-computer-vision-from-pixel-to-detection-to-gen-cv/learn/lecture/54776769#overview

AI vs ML vs Deep Learning vs Generative AI

AI:

- What is Artificial intelligence ?
  - This is the one of the trickiest question of history.Even the very famous person named Marvin Minsky , In the year 1958 he released a research paper that says :

> I feel that it would not be useful to lay down any absolute definition of "intelligence" or of "intelligent behaviour". For our goals in trying to design "thinking machines" are constantly changing in relation to our ever-increasing resources in this area.

  - It is not possible to get a definition of ai
  - AGI : Artificial general intelligence , the strong AI and it is hypothetical
  - And we have Narrow Ai also called as weak AI and also perform well can outperform human but they are limited to a field

ML :

- The most popular way of using Artificial intelligence is using machine learning .
- But theses terms are different :
  - Ai is an umbrella term which we defines as “all the various techniques you used to make computer do smart things”
  - And ML is one of the way to do these AI task but it is not just one of the way to do it .
- Machine learning is a **subset** of AI where the machine tries and learn from examples instead of programming explicit rules.
- Traditional techniques for Ai is good but only when you know all the rules properly but ML is not like that it gathers information from provided data to learn from
- Machine need huge amount of data to gather from and that is why the terms like big data is so popular in the field

Deep Learning :

- Deep learning is a smaller subset of ML
- It is actually a machine learning that uses neural networks that is like something like brain

- This is not exactly human instead they are inspired by human brain , just like our brain has billions of neurons and synaptic connections that is a connection between neurons
- Neural network are somewhat like simulated neurons where individual nodes are connected and can send messages to other stimulated neurons
- And it actually works with layers of neurons
- Advantages :
    - Gives better results in Complex scenario like :
        - Image and video processing ,
        - language recognition ,
        - complex game systems ,
        - recommendation systems ,
        - medical diagnoses
    - And mostly all the genAi models uses DL
- Disadvantages :
    - Expensive and Demanding : Computationally expensive , requires a lot of data , takes a lot of time , and often expensive
- Exactly how much do we need model to train the ml model and this is situation dependent and it is not usual to get thousand and millions information in order to train dl model and that is why its very costly

GenAI

- As the name says it is the AI that can generate and create new and original content such as text , images , music , video and computer code.
- For example for generating images they are trained on large amount of images and by using the trained model and the prompt giving they generates their brand new data.
- Genai model uses LLM to generate information, for text based searching and generation
- LLM : An Ai system extensively trained on vast amounts of text data, to interact with and generate human languages .
- And for other features we can use foundation models and LLM is one of them
- Foundation models : An Ai Trained on huge amount odf data , to then adapt to multiple uses , Foundation models include text-based LLms and image -based generative AI
- Hallucination : Generative AI may create Results and answers that look convincing but are entirely invented with no basis in reality
- Genai can help us generate new content but that doesn’t really specify that the result generated is correct

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)	Generative AI (GenAI)
Definition	The broad field of creating computer systems that perform tasks requiring human-like intelligence.	A subset of AI in which systems learn patterns from data and examples instead of relying entirely on explicitly programmed rules.	A subset of ML that uses multi-layered neural networks to learn complex patterns from large amounts of data.	A branch of AI that generates new content such as text, images, audio, video, and computer code.
Relationship	The broadest concept. AI includes ML, DL, rule-based systems, search algorithms, expert systems, and other techniques.	ML is one approach used to build AI systems.	DL is a specialized approach within ML.	GenAI is defined by what the system does—generates new content—and commonly relies on DL and foundation models.
Main objective	To make machines perform tasks that appear intelligent.	To enable machines to learn from data and make predictions or decisions.	To solve complex problems by automatically learning hierarchical representations from data.	To produce new, human-like content based on patterns learned during training.
How it works	It may use explicitly programmed rules, logic, search, optimization, ML, or DL.	The model identifies patterns in training data and uses them to make predictions or decisions on new data.	The model processes data through multiple interconnected neural-network layers, progressively learning simple-	The model learns patterns from large datasets and uses a prompt or input to generate new content.

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)	Generative AI (GenAI)
			to-complex features.	
Type of output	Decisions, classifications, predictions, recommendations, actions, or automated responses.	Predictions, classifications, scores, recommendations, or decisions.	Complex predictions, classifications, generated representations, or learned actions.	Newly generated text, images, music, video, speech, or code.
Examples	Expert systems, rule-based decision systems, search systems, recommendation systems, autonomous systems, and intelligent assistants.	Spam detection, house-price prediction, fraud detection, customer churn prediction, and credit-risk assessment.	Image recognition, video analysis, speech recognition, medical-image analysis, complex recommendation systems, and game-playing systems.	Chatbots, text generation, image generation, code generation, music generation, video generation, and document summarization.
Human-brain connection	AI is inspired by the goal of reproducing or simulating intelligent behavior, but it is not necessarily modeled on the human brain.	ML is not required to imitate the brain; it focuses on learning patterns from data.	Neural networks are loosely inspired by biological neurons and their connections, but they are mathematical models and are not exact copies of the human brain.	GenAI models learn patterns from data and do not possess human understanding, consciousness, or guaranteed factual knowledge.

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)	Generative AI (GenAI)
Major advantage	Provides a general framework for automating intelligent tasks across many domains.	Can discover patterns and make useful predictions without manually defining every rule.	Performs particularly well on complex tasks involving images, speech, video, language, and other unstructured data.	Can produce new content quickly and support tasks such as writing, coding, design, summarization, and content creation.
Major limitation	The meaning and scope of AI are broad, and an AI system may be intelligent only within a specific task or domain.	Performance depends on data quality, task design, feature representation, and the suitability of the chosen algorithm.	Often computationally expensive, data-intensive, time-consuming to train, and costly to deploy.	It may produce convincing but incorrect or invented information, commonly called a hallucination.
Accuracy and reliability	Depends on the underlying technique and the quality of its design and data.	Usually evaluated using task-specific metrics such as accuracy, precision, recall, or mean squared error.	Can achieve excellent results, but high performance does not guarantee explainability or correctness in every situation.	The generated output may be fluent and realistic without being factually correct, so it requires verification.
Common technologies	Rule engines, expert systems, search algorithms, optimization methods, ML, and DL.	Decision trees, linear regression, logistic regression, support-vector machines, clustering, and neural networks.	Convolutional neural networks, recurrent neural networks, transformers, and other deep neural-network architectures.	Large language models, diffusion models, multimodal models, and other

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)	Generative AI (GenAI)
				foundation models.
Factual correctness	Depends on the system, its rules, data, and validation process.	Predictions depend on the quality and relevance of the training data.	Complex models can produce highly accurate results but may still make errors.	GenAI is optimized to generate plausible content, not automatically to guarantee truth or correctness.

Source(Pluralsight) : https://app.pluralsight.com/library/courses/artificial-intelligence-explained

Real World CV use Cases and training outcomes

1. **Medical image analysis**
   a. **How is Computer Vision used :**Computer vision analyzes X-rays, CT scans, MRI scans, ultrasound images, and retinal images to identify abnormalities such as tumors, fractures, or signs of disease. It can also highlight the affected region for further examination.
   b. **Training outcome :** After training, a learner can prepare and preprocess medical-image datasets, classify images, detect suspicious regions, apply image segmentation, evaluate model performance, and understand why medical predictions must be reviewed by qualified healthcare professionals.
2. **Manufacturing quality inspection**
   a. **How is Computer Vision used:** Cameras inspect products on an assembly line and identify scratches, cracks, missing parts, incorrect assembly, or other defects.
   b. **Training outcome :** After training, a learner can collect and label product images, train a defect-detection model, classify products as defective or non-defective, detect missing components, and design a visual inspection pipeline for production environments.

Manufacturing is a major application of computer vision for identifying defective products and monitoring production quality.

3. **Retail inventory and checkout automation**
    a. **How is computer vision used :** Vision systems monitor shelves, count products, detect empty spaces, identify misplaced items, and recognize products during automated checkout.
    b. **Training Outcome :** After training, a learner can build a product-detection system, count objects in images or video, monitor shelf availability, recognize labels or barcodes, and create a basic inventory-monitoring application. Retailers use visual recognition for inventory management and customer-related analysis.
4. **Agricultural crop and disease monitoring:**
    a. **How is Computer Vision used :** Drones, satellites, or field cameras capture crop images. Computer vision identifies unhealthy plants, weeds, pests, disease symptoms, and crops that are ready for harvesting.
    b. **Training Outcome :** After training, a learner can process field images, classify healthy and unhealthy crops, detect weeds or diseased regions, estimate crop counts, and develop a monitoring system to support targeted agricultural treatment. Computer vision is used to monitor crop disease, insect infestations, and crop readiness.

Source : https://www.ibm.com/think/topics/artificial-intelligence-business-use-cases